INS 401: Project Management

Managing Project Resources

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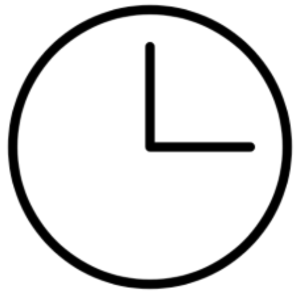
1 December 2025
Main Campus, Lagos

Ground Rules

- Participate
- Respect each other
- Silence Phones
- Be on time



Outline

- Types of resources
- Resource allocation and levelling techniques
- Strategies for managing constrained resources

Learning Outcomes

At the end of this class, students should be able to:

- describe the various types of project resources
- discuss resource allocation and levelling techniques
- discuss various strategies for managing constrained resources

Types of Project Resources

In project management, resources refer to the essential elements, both tangible and intangible, required to ensure successful completion of projects.

Proper allocation, monitoring and control of the various types of resources is crucial to realizing the intended outcomes of any project.

Project resources may be classified under the following four categories:

- Human Resources
- Physical Resources
- Financial Resources
- Time

Human Resources

- The key human resources here are the project team members.
- The project team, led by the project manager, consists of individuals with assigned roles and responsibilities who work collectively to achieve a shared project goal.
- The project team may include full-time and part-time members, as well as experts with specific skills who may be brought in as needed.

Implications

- Limited availability of skilled and competent personnel, their overallocation to tasks, and insufficient training may result in delays in project delivery, personnel burnout and poor quality of work.

Physical Resources

- Physical resources include equipment, materials, facilities and infrastructure.
- Effective physical resource allocation and management is crucial for successful completion of projects.

Implications

- Failure to secure critical equipment or infrastructure on time may result in project delay.
- Ordering low quality material may cause a high rate of recalls or rework.
- Keeping too much inventory may result in high operations cost and reduce the organization’s profit; too low inventory may cause material shortages that could result to delays in project delivery.

Financial Resources

- The critical issue in financial resources is the allocation of funds for various project components such as labour, equipment, materials and overhead costs.

- Proper budgeting, tracking of costs and cost control are essential for efficient and successful completion of projects.

Implications

- Insufficient funding to cover project costs may result in quality issues, delay of project deliverables or outright cancellation of the project.

- Funding problems may arise due to poor budgeting, unforeseen expenses or changes in scope.

- Over-expenditure and uncontrolled budget spending may lead to budget overrun, which might affect the viability of the project.

Time as a Project Resource

- Time is considered a critical resource in project management, as it determines project schedules and deadlines.

- Through accurate estimation of the time required for various tasks, project managers can allocate resources more effectively, ensuring that the project remains on schedule and within budget.

- Real-time tracking of projects help in identifying potential bottlenecks and reallocating resources as necessary to keep the project on track.

Implications

- Unrealistic deadlines, unclear timelines, or unexpected delays may lead to rushed work, quality issues and failure to meet stakeholder expectations.

Project Resource Allocation

- Key processes that relate to project resource allocation are the following:

  - Estimate Activity Resources
  - Acquire Resources
  - Estimate Activity Durations
  - Develop Schedule
  - Estimate Costs
  - Determine Budget

Estimate Activity Resources

Estimate Activity Resources is the process of estimating team resources and the type and quantities of material, equipment, and supplies necessary to complete the work.

Core outputs

- Resource Requirements
- Basis of Estimates
- Resource Breakdown Structure.

Estimate Activity Resources

Inputs

1. Project management plan
   -Resource management plan
   -Scope baseline
2. Project documents
   -Activity list
   -Activity attributes
   -Assumption log
   -Cost estimates
   -Resource calendars
   -Risk register
3. Enterprise environmental factors
4. Organizational process assets

Tools & Techniques

1. Expert judgement
2. Bottom-up estimating
3. Analogous estimating
4. Parametric estimating
5. Data analysis
   -Alternative analysis
6. Project management information system
7. Meetings

Outputs

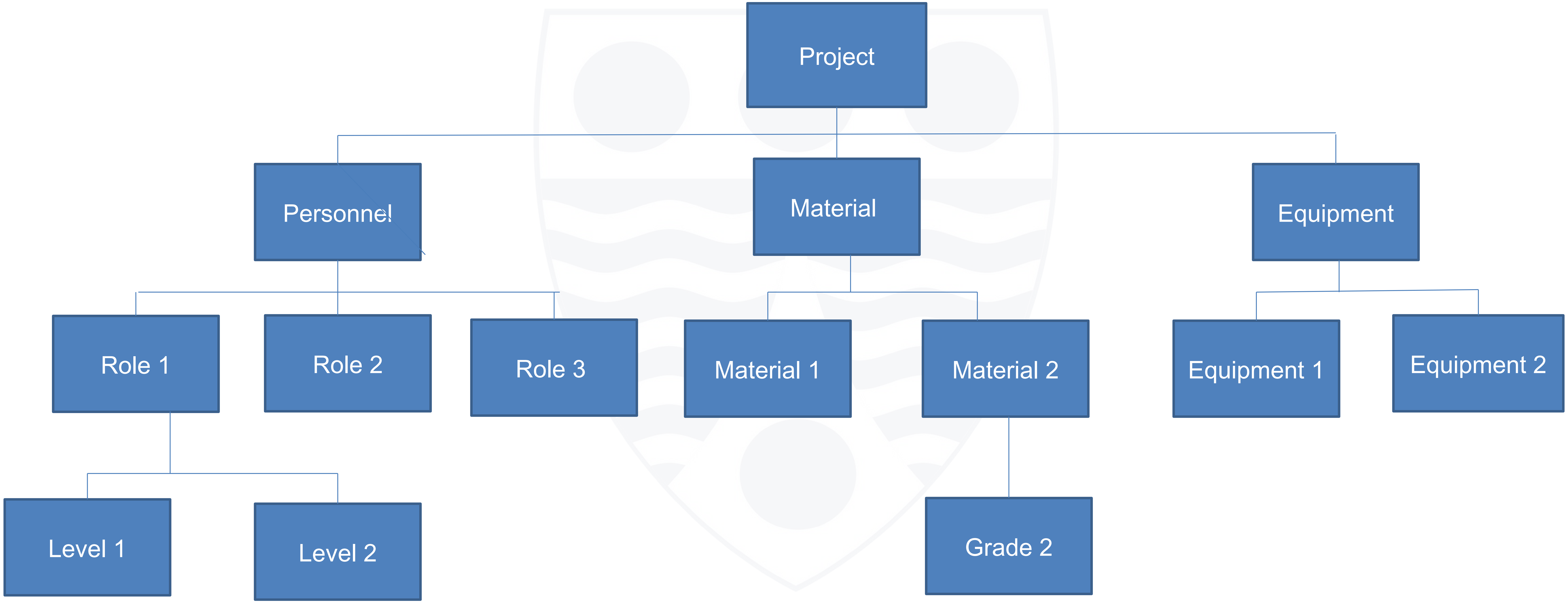
1. Resource requirements
2. Basis of estimates
3. Resource breakdown structure
4. Project documents updates
   -Activity attributes
   -Assumption log
   -Lessons learned register

Resource Requirements and Resource Breakdown Structure [RBS]

Resource Requirements: The types and quantities of resources required for each work package or activity in a work package and can be aggregated to determine the estimated resources for each work package, each WBS branch, and the project as a whole.

Resource Breakdown Structure (RBS): A hierarchical representation of resources by category and type. Examples of resource categories include labour, material, equipment and supplies.

Resource Breakdown Structure [Sample]



Acquire Resources

Acquire Resources is the process of obtaining team members, facilities, equipment, materials, supplies, and other resources necessary to complete project work.

Key Outputs

Physical Resource Assignments: Documentation of the physical resource assignments, which records the materials, equipment, supplies, locations and other physical resources that will be used during the project.

Project Team Assignments: Documentation that identifies the team and member roles and responsibilities. The documentation may include a project team directory and names inserted into the project organization charts and schedules.

Resource Calendar: A calendar that identifies the working days and shifts upon which each specific resource is available. Time zones and expertise of personnel are important considerations here.

Acquire Resources

Inputs

1. Project management plan
   - -Resource management plan
   - -Procurement management plan
   - -Cost baseline
2. Project documents
   - -Project schedule
   - -Resource calendars
   - -Resource requirements
   - -Stakeholder register
3. Enterprise environmental factors
4. Organizational process assets

Tools & Techniques

1. Decision making
   - -Multicriteria decision analysis
2. Interpersonal and team skills
   - -Negotiation
3. Pre-assignment
4. Virtual teams

Outputs

1. Physical resource assignments
2. Project team assignment
3. Resource calendars
4. Change requests
5. Project management plan updates
   - -Resource management plan
   - -Cost baseline
6. Project documents updates
   - -Resource breakdown structure
   - -Resource requirements
   - -Project schedule
   - -Risk register
   - -Lessons learned register
7. Enterprise environmental factors
8. Organizational process assets

Estimate Activity Durations

Estimate activity durations is the process of estimating the number of work periods needed to complete individual activities with estimated resources. It provides the amount of time each activity will take to complete.

Key Outputs

- **Duration estimates** – these are quantitative assessments of the likely number of time periods that are required to complete an activity.
- **Basis of estimates** – a supporting documentation that provides a clear understanding of how the duration estimate was derived.

Estimate Activity Durations

Inputs	Tools & Techniques	Outputs
1. Project management plan -Schedule management plan -Scope baseline 2. Project documents -Activity attributes -Activity list -Assumption log -Lessons learned register -Milestone list -Project team assignments -Resource breakdown structure -Resource calendars -Resource requirements -Risk register 3. Enterprise Environmental factors 4. Organizational process assets	1. Expert judgement 2. Analogous estimating 3. Parametric estimating 4. Three-point estimating 5. Bottom-up estimating 6. Data analysis -Alternative analysis -Reserve analysis 7. Decision making 8. Meetings	1. Duration estimates 2. Basis of estimates 3. Project documents updates -Activity attributes -Assumption log -Lessons learned register

Develop Schedule

Inputs

1. Project management plan
   - -Schedule management plan
   - -Scope baseline
2. Project documents
   - -Activity attributes
   - -Activity list
   - -Assumption log
   - -Lessons learned register
   - -Milestone list
   - -Project team assignments
   - -Project schedule network diagrams
   - -Resource calendars
   - -Resource requirements
   - -Risk register
3. Enterprise Environmental factors
4. Organizational process assets

Tools & Techniques

1. Schedule network analysis
2. Critical path method
3. Resource optimization
4. Data analysis
   - -What-if scenario analysis
   - -Simulation
5. Leads and lags
6. Schedule compression
7. Project management information system
8. Agile release planning

Outputs

1. Schedule baseline
2. Project schedule
3. Schedule data
4. Project calendars
5. Change request
6. Project management plan updates
   - -Schedule management plan
   - -Cost baseline
7. Project documents updates
   - -Activity attributes
   - -Assumption log
   - -Duration estimates
   - -Lessons learned register
   - -Resource requirements
   - -Risk register

Estimate Costs

Estimate costs is the process of developing an approximation of the cost of resources needed to complete project work. It determines the monetary resources required for the project.

Key Outputs

- **Cost estimates** – these are quantitative assessments of the likely costs for resources required to complete an activity or the project. Costs are estimated for all resources that will be charged to the project, including labour, materials, equipment, services and facilities, as well as other things like cost of financing, allowance for inflation and exchange rates.

- **Basis of estimates** – a supporting documentation that provides a clear understanding of how the cost estimates were derived.

PMI Process Groups and Knowledge Areas Mapping

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PROCESS GROUP AND KNOWLEDGE AREA MAPPING

Knowledge Areas	Project Management Process Groups				
	Initiating	**Planning**	**Executing**	**Monitoring and Controlling**	**Closing**
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Source: *PMBOK® Guide*, 6th Edition

Estimate Costs

Inputs

1. Project management plan
   - -Cost management plan
   - -Quality management plan
   - -Scope baseline
2. Project documents
   - -Lessons learned register
   - -Project schedule
   - -Project team assignments
   - -Resource requirements
   - -Risk register
3. Enterprise Environmental factors
4. Organizational process assets

Tools & Techniques

1. Expert judgement
2. Analogous estimating
3. Parametric estimating
4. Three-point estimating
5. Bottom-up estimating
6. Data analysis
   - -Alternative analysis
   - -Reserve analysis
   - -Cost of quality
7. Project management information system
8. Decision making
   - -Voting

Outputs

1. Cost estimates
2. Basis of estimates
3. Project documents updates
   - -Assumption log
   - -Lessons learned register
   - -Risk register

Determine Budget

Determine budget is the process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline. It determines the cost baseline against which project performance can be monitored and controlled.

Key Outputs

- **Cost baseline** – this is the approved version of the time-phased project budget, excluding any management reserves, which can only be changed through formal change control procedures.
- **Project funding requirements** – Forecast project costs to be paid that are derived from the cost baseline for total or periodic requirements, including projected expenditures plus anticipated liabilities.

Project Budget = Cost Baseline + Management Reserve

Determine Budget

Inputs

1. Project management plan
   - -Cost management plan
   - -Resource management plan
   - -Scope baseline
2. Project documents
   - -Lessons learned register
   - -Cost estimates
   - -Project schedule
   - -Risk register
3. Business documents
   - -Business case
   - -Benefits management plan
4. Enterprise Environmental factors
5. Organizational process assets

Tools & Techniques

1. Expert judgement
2. Cost aggregation
3. Data analysis
   - -Reserve analysis
4. Historical information review
5. Funding limit reconciliation
6. Financing

Outputs

1. Cost baseline
2. Project funding requirements
3. Project documents updates
   - -Cost estimates
   - -Project schedule
   - -Risk register

Strategies for Managing Constrained Resources

Managing constrained resources, such as insufficient personnel, tight project deadlines and budget, could be very challenging.

Two key strategies for managing resource constraints are discussed here:

Resource levelling – a technique in which start and finish dates are adjusted based on resource constraints with the goal of balancing the demand for resources with the available supply. It can be used when shared or critically required resources are available only at certain times or in limited quantities, or are over-allocated. Resource levelling may cause the original critical path to change.

Resource smoothing – a technique that adjusts the activities of a schedule model such that the requirements for resources on the project do not exceed certain predefined resource limits. It involves adjusting activities within the float of the project to avoid overallocation without changing the project duration or affecting the critical path. Resource Smoothing is performed to achieve a more uniform resource utilization over a period of time.

Resource Levelling vs. Resource Smoothing

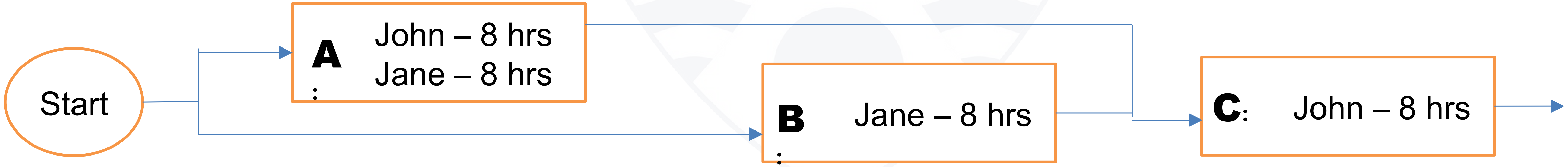
Resource Levelling	Resource Smoothing
Objective is to balance resource demand with supply	Objective is to achieve uniform resource utilization
Often causes the critical path to change	The project's critical path is not changed. Utilizes activity float to smoothen resource demand
Critical resources are the primary constraint	Project duration is the primary constraint
It is performed when key resources are only available at certain times, or in limited quantities, or over allocated	The reasons to perform it are limits on resource usage, or uneven resource utilization
It allows you to change your project's start and end dates when you have limited resources	It involves adjusting your resource requirements in order to meet the initial deadline

Resource Levelling

Before Resource Levelling

Day 1	Day 2	Day 3
John: 8 hrs; Jane: 16 hrs	Jane: 8 hrs	

After Resource Levelling



Day 1	Day 2	Day 3
John: 8 hrs; Jane: 8 hrs	Jane: 8 hrs	John: 8 hrs

Resource Smoothing

Before Resource Smoothing

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
Activity A Resources	**4**	**5**	**5**	**5**	**5**	**5**
Activity B Resources		1	2	2	Free float	Free float

After Resource Smoothing

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
Activity A Resources	**4**	**5**	**5**	**5**	**5**	**5**
Activity B Resources		1	1	1	1	1

Constraints:

1. Only 5 resources are available to work on activity A at any point (scarce resources)
2. A maximum of 6 resources are to be assigned to the project at any point in time (predefined resource limit)

Conclusion

- Proper allocation, monitoring and control of the various types of resources is crucial to realizing the intended outcomes of any project.

- Resource levelling and resource smoothing are two resource optimization techniques that project managers can use to manage resource constraints.

Closing Remarks